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Semiconductor device

Abstract

A device includes a semiconductor fin semiconductor fin extending from a substrate, a gate structure extending across the semiconductor fin, and a multilayer gate spacer on a sidewall of the gate structure. The multilayer gate spacer includes an inner spacer layer, an outer spacer layer, and a dielectric structure. The inner spacer layer has a vertical portion extending along the sidewall of the gate structure, and a lateral portion laterally extending from the vertical portion in a direction away from the gate structure. The outer spacer layer is spaced apart from the vertical portion of the inner spacer layer by an air gap. The dielectric structure spaces apart a bottom end of the outer spacer layer from the lateral portion of the inner spacer layer.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6001695	12/1998	Wu	438/307	H10D 64/679
8962400	12/2014	Tsai et al.	N/A	N/A
9093514	12/2014	Tsai et al.	N/A	N/A
9236267	12/2015	De et al.	N/A	N/A
9245805	12/2015	Yeh et al.	N/A	N/A
9418897	12/2015	Ching et al.	N/A	N/A
9520482	12/2015	Chang et al.	N/A	N/A
9576814	12/2016	Wu et al.	N/A	N/A
9608065	12/2016	Bergendahl et al.	N/A	N/A
9608116	12/2016	Ching et al.	N/A	N/A
9666533	12/2016	Basker et al.	N/A	N/A
9812363	12/2016	Liao et al.	N/A	N/A
9859380	12/2017	Lee et al.	N/A	N/A
10026824	12/2017	Chanemougame et al.	N/A	N/A
10367076	12/2018	Cheng et al.	N/A	N/A
2005/0189660	12/2004	Bu	257/900	H10D 30/0227
2010/0213554	12/2009	Wang	257/E21.214	H01L 29/6656
2014/0110798	12/2013	Cai et al.	N/A	N/A
2016/0163816	12/2015	Yu et al.	N/A	N/A
2016/0181143	12/2015	Kwon et al.	N/A	N/A
2016/0284806	12/2015	Park et al.	N/A	N/A
2016/0365426	12/2015	Ching et al.	N/A	N/A
2017/0162650	12/2016	Cheng et al.	N/A	N/A
2017/0194423	12/2016	Lin	N/A	N/A

2018/0138280	12/2017	Li et al.	N/A	N/A
2018/0182845	12/2017	Seong et al.	N/A	N/A
2018/0233398	12/2017	Van Cleemput	N/A	H01L 21/67259
2018/0269301	12/2017	Cheng et al.	N/A	N/A
2018/0323276	12/2017	Kang et al.	N/A	N/A
2018/0374927	12/2017	Liu et al.	N/A	N/A
2019/0043959	12/2018	Lee et al.	N/A	N/A
2019/0148501	12/2018	Chen et al.	N/A	N/A
2019/0157414	12/2018	Ando et al.	N/A	N/A
2019/0165121	12/2018	Park	N/A	N/A
2019/0296123	12/2018	Lee et al.	N/A	N/A
2019/0334008	12/2018	Chen et al.	N/A	N/A
2019/0393335	12/2018	Economikos et al.	N/A	N/A
2020/0027959	12/2019	Cheng et al.	N/A	N/A
2020/0127109	12/2019	Wang et al.	N/A	N/A
2020/0219989	12/2019	Cheng	N/A	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is continuation of U.S. application Ser. No. 17/129,253, filed Dec. 21, 2020, which is continuation of U.S. application Ser. No. 16/914,940, filed Jun. 29, 2020, now U.S. Pat. No. 10,872,965, issued Dec. 22, 2020, which is a divisional of U.S. application Ser. No. 16/047,038, filed Jul. 27, 2018, now U.S. Pat. No. 10,700,180, issued Jun. 30, 2020, all of which are incorporated herein by reference in their entirety.

BACKGROUND

(1) As the semiconductor industry has strived for higher performance, higher device density, and lower costs, problems involving both fabrication and design have been encountered. One solution to these problems has been the development of a fin-like field effect transistor (FinFET). A FinFET includes a thin vertical ‘fin’ formed in a free standing manner over a major surface of a substrate. The source, drain, and channel regions are defined within this fin. The gate in the transistor crosses over the channel region of the fin. This configuration allows the gate to induce current flow in the channel from three sides. Thus, FinFET devices have the advantage of higher current flow and reduced short-channel effects.

(2) The dimensions of FinFETs and other metal oxide semiconductor field effect transistors (MOSFETs) have been reduced as technological advances have been made in integrated circuit materials. For example, high-k metal gate (HKMG) processes have been applied to FinFETs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 is a flowchart of a method for manufacturing a semiconductor structure, in accordance with some embodiments of the instant disclosure.

(3) FIGS. 2-14 are perspective views illustrating intermediate stages in the manufacturing of a semiconductor structure, in accordance with some embodiments of the instant disclosure.

(4) FIG. 15A is a cross-sectional view along a line A-A' in FIG. 14.

(5) FIG. 15B is a cross-sectional view along a line B-B' in FIG. 14.

(6) FIGS. 16A, 16B, 17A, 17B, 18, 19A, 19B, 20A, 20B, 21, 22A, 22B, 23A, 23B, and 24 are cross-sectional views illustrating intermediate stages in the manufacturing of a semiconductor structure, in accordance with some embodiments of the instant disclosure.

DETAILED DESCRIPTION

(7) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(8) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(9) The fins may be patterned by any suitable method. For example, the fins may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins.

(10) The instant disclosure provides a method for manufacturing a semiconductor structure. FIG. 1 is a flowchart of a method **100** for manufacturing a semiconductor structure, in accordance with some embodiments of the instant disclosure. Operation **101** of the method is forming a semiconductor substrate having a fin. The method continues with operation **103** in which a dummy gate structure is formed to cross over the fin. Operation **105**, gate spacers are formed adjacent to the dummy gate structure. The method continues with operation **107** in which a source and a drain are formed adjacent to the gate spacers. The method continues with operation **109** in which a contact etch stop layer and an interlayer dielectric layer are formed adjacent to the gate spacers. The method continues with operation **111** in which the dummy gate structure is replaced with a gate structure. Operation **113**, at least one portion of one of the gate spacers is removed. The method continues with operation **115** in which a contact stop layer is formed on the gate structure, the gate spacers, and the interlayer dielectric layer to form an air gap. It is understood that FIG. 1 has been simplified for a good understanding of the concepts of the instant disclosure. Accordingly,

it should be noted that additional processes may be provided before, during, and after the methods of FIG. 1, and that some other processes may only be briefly described herein.

(11) FIGS. 2-14 are perspective views illustrating intermediate stages in the manufacturing of a semiconductor structure, in accordance with some embodiments of the instant disclosure.

(12) In the operation 101 of FIG. 1, a semiconductor substrate having a fin is formed. Please refer to FIGS. 2-6. As shown in FIG. 2, a semiconductor substrate 210 is illustrated. In some embodiments, the semiconductor substrate 210 is a bulk semiconductor, a semiconductor-on-insulator (SOI) substrate, or the like. For example, the substrate 210 may be a wafer, such as a silicon wafer. Generally, an SOI substrate includes an insulator layer and a layer of a semiconductor material thereon. For example, the insulator layer may be a buried oxide (BOX) layer, a silicon oxide layer, or the like. The insulator layer is provided on a substrate, a silicon, or a glass substrate. Other substrates, such as a multi-layered or gradient substrate may also be used. In some embodiments, the semiconductor material of the semiconductor substrate 210 may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof.

(13) Still referring to FIG. 2, a pad layer 220 is formed on the semiconductor substrate 210, a mask layer 230 is formed on the substrate 210, and a photo-sensitive layer 240 is formed on the mask layer 230. The pad layer 220 may act as an adhesion layer between the substrate 210 and mask layer 230, and may also act as an etch stop layer for etching the mask layer 230. The mask layer 230 may be used as a hard mask during photolithography processes. In some embodiments, the pad layer 220 is formed using thermal oxidation process. For example, the pad layer 220 is a thin film including silicon oxide formed by a thermal oxidation process. In some embodiments, the mask layer 230 is formed using low-pressure chemical vapor deposition (LPCVD) or plasma enhanced chemical vapor deposition (PECVD). For example, the mask layer 230 includes silicon nitride.

(14) As shown in FIG. 3, portions of the mask layer 230 and the pad layer 220 not covered and protected by the photo-sensitive layer 240 are etched to expose the semiconductor substrate 210, and the semiconductor substrate 210 is then etched to form a fin 250 of the semiconductor substrate 210. Next, the photo-sensitive layer 240 may be removed. A cleaning step may be optionally performed to remove a native oxide of the semiconductor substrate 210. The cleaning may be performed using diluted hydrofluoric (HF) acid, for example.

(15) Please refer to FIG. 4. An isolation dielectric layer 260 is formed to cover the mask layer 230, the pad layer 220, and the fin 250. In some embodiments, the isolation dielectric layer 260 is formed using a high-density-plasma (HDP) chemical vapor deposition (CVD) process, a sub-atmospheric CVD (SACVD) process, a high aspect-ratio process (HARP), or spin-on-dielectric (SOD) process. For example, the isolation dielectric layer 260 includes silicon oxide, silicon nitride, silicon oxynitride, fluoride-doped silicate glass (FSG), or other low-K dielectric materials. Other processes and materials may be used.

(16) As shown in FIG. 5, a planarization process is performed to remove a portion of the isolation dielectric layer 260, the mask layer 230, and the pad layer 220 to expose the fin 250. For example, the planarization process is a chemical mechanical polish (CMP) process. In embodiments, the planarization process is performed to remove a portion of the isolation dielectric layer 260 to expose the mask layer 230, which may act as the CMP stop layer. Subsequently, the mask layer 230, if formed of silicon nitride, may be removed by a wet process using hot H₂PO₄, and the pad layer 220, if formed of silicon oxide, may be removed using diluted HF, according to some examples of the present disclosure.

(17) Please refer to FIG. 6. The isolation dielectric layer 260 is recessed, for example, through an etching operation, wherein diluted HF, SiCoNi (including HF and NH₃), or the like, may be

used as the etchant. In some embodiments, the isolation dielectric layer **260** is referred to as shallow trench isolation (STI) structure. After recessing the isolation dielectric layer **260**, a top surface of the fin **250** is higher than a top surface of the isolation dielectric layer **260**. Therefore, an upper portion of the fin **250** protrudes above the isolation dielectric layer **260**, and a lower portion of the fin **250** is embedded in the isolation dielectric layer **260**.

(18) In the operation **103** of FIG. **1**, a dummy gate structure is formed to cross over the fin. Reference is made to FIGS. **7-9**. As shown in FIG. **7**, a gate oxide layer **270** is formed to wrap the fin **250**, and portions of the isolation dielectric layer **260** are not covered by the gate oxide layer **270**. In some embodiments, the gate oxide layer **270** is formed using a deposition processes, such as chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), plasma enhanced CVD (PECVD) or other suitable techniques. In some embodiments, the gate oxide layer **270** is made of high-k dielectric materials, such as metal oxides, transition metal-oxides, or the like. Examples of the high-k dielectric material include, but are not limited to, hafnium oxide (HfO₂), hafnium silicon oxide (HfSiO), hafnium tantalum oxide (HfTaO), hafnium titanium oxide (HfTiO), hafnium zirconium oxide (HfZrO), zirconium oxide (ZrO₂), zirconium silicon oxide (ZrSiO₄), lanthanum oxide (La₂O₃), titanium oxide (TiO₂), tantalum pentoxide (Ta₂O₅), strontium titanium oxide (SrTiO₃), aluminum oxide (Al₂O₃), hafnium dioxide-alumina (HfO₂—Al₂O₃) alloy, scandium oxide (Sc₂O₃), barium strontium titanium oxide (Ba_{0.6}Sr_{0.4}TiO₃), yttrium oxide (Y₂O₃), a combination thereof, or other suitable dielectric materials. The gate oxide layer **270** may include various dielectric materials, either individually or in combination.

(19) Please refer to FIG. **8**. After the gate oxide layer **270** is formed, a dummy gate electrode layer **280** is subsequently formed over the gate oxide layer **270**, and a mask layer **290** is formed on the dummy gate electrode layer **280**. In some embodiments, the dummy gate electrode layer **280** is deposited by CVD, PVD, sputter deposition, or other techniques suitable for depositing conductive materials. In some embodiments, the dummy gate electrode layer **280** includes polycrystalline-silicon (poly-Si), polycrystalline silicon-germanium (poly-SiGe), metallic nitrides, metallic silicides, metallic oxides, or metals. In some embodiments, the dummy gate electrode layer **280** includes a metal-containing material such as TiN, TaN, TaC, Co, Ru, Al, combinations thereof, or multi-layers thereof. The mask layer **290** may be hard masks for protecting the underlying dummy gate electrode layer **280** against subsequent etching process. The mask layer **290** may be formed by a series of operations including deposition, photolithography patterning, and etching processes. The photolithography patterning processes may include photoresist coating (e.g., spin-on coating), soft baking, mask aligning, exposure, post-exposure baking, developing the photoresist, rinsing, drying (e.g., hard baking), and/or other applicable processes. The etching processes may include dry etching, wet etching, and/or other etching methods (e.g., reactive ion etching).

(20) The dummy gate electrode layer **280** is then etched, using the mask layer **290** as an etching mask, to form a dummy gate structure DG crossing over the fin **250** of the semiconductor substrate **210**, as shown in FIG. **9**. More specifically, a portion of the fin **250** is wrapped by the dummy gate structure DG. The gate oxide layer **270** is disposed between the fin **250** and the dummy gate structure DG. In some embodiments, the dummy gate structure DG has a longitudinal axis substantially perpendicular to a longitudinal axis of the fin **250**. In some embodiments, the dummy gate structure DG may be later replaced with a replacement gate structure using a “gate-last” or replacement-gate process.

(21) In the operation **105** of FIG. **1**, gate spacers are formed adjacent to the dummy gate structure. Attention is now invited to FIGS. **10-11**. In embodiments, as shown in FIG. **10**, a gate spacer layer GSL1 is blanket formed over the dummy gate structure DG, the fin **250**, and the isolation dielectric layer **260**, a gate spacer layer GSL2 is formed on the gate spacer layer GSL1, and a gate spacer layer GSL3 is formed on the gate spacer layer GSL2. In some embodiments, the gate spacer layer GSL1 is conformally formed on the dummy gate structure DG, the fin **250**, and the isolation dielectric

layer **260**. The gate spacer layer **GSL2** is conformally formed on the gate spacer layer **GSL1**. The gate spacer layer **GSL3** is conformally formed on the gate spacer layer **GSL2**. For example, the gate spacer layer **GSL1**, the gate spacer layer **GSL2**, and the gate spacer layer **GSL3** independently include silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, silicon carbonitride, silicon oxycarbonitride, silicon oxycarbide, porous dielectric materials, hydrogen doped silicon oxycarbide (SiOC:H), low-k dielectric materials, or other suitable dielectric material. In some embodiments, the gate spacer layer **GSL2** includes a material different from that of the gate spacer layer **GSL1** and/or the gate spacer layer **GSL3**. In some embodiments, the gate spacer layer **GSL2** has an etch resistance different from that of the gate spacer layer **GSL1** and/or the gate spacer layer **GSL3**, with respect to an identical etchant. In some embodiments, the gate spacer layer **GSL2** has an etch resistance lower than that of the gate spacer layer **GSL1** and/or the gate spacer layer **GSL3**, with respect to an identical etchant. In some embodiments, the gate spacer layer **GSL1**, the gate spacer layer **GSL2**, and the gate spacer layer **GSL3** respectively have a thickness ranged from about 5 Å to about 1000 Å.

(22) Please refer to FIG. **11**. In some embodiments, the gate spacer layer **GSL1**, the gate spacer layer **GSL2**, the gate spacer layer **GSL3**, the gate oxide layer **270** are etched to expose a portion of the fin **250** and the isolation dielectric layer **260**, thereby forming a gate spacer **GS1** adjacent to the dummy gate structure **DG**, a gate spacer **GS2** adjacent to the gate spacer **GS1**, and a gate spacer **GS3** adjacent to the gate spacer **GS2**. As such, the gate spacer **GS1** is formed to cover a sidewall **DSW** of the dummy gate structure **DG**. The gate spacer **GS2** is formed on the gate spacer **GS1**. The gate spacer **GS3** is formed on the gate spacer **GS2**. In examples, the gate spacer **GS1**, the gate spacer **GS2**, and the gate spacer **GS3** independently include silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, silicon carbonitride, silicon oxycarbonitride, silicon oxycarbide, porous dielectric materials, hydrogen doped silicon oxycarbide (SiOC:H), low-k dielectric materials, or other suitable dielectric material. In some embodiments, the gate spacer **GS2** includes a material different from that of the gate spacer **GS1** and/or the gate spacer **GS3**. In yet some embodiments, the gate spacer **GS2** has an etch resistance different from that of the gate spacer **GS1** or the gate spacer **GS3**. In some embodiments, the gate spacer **GS2** has an etch resistance lower than that of the gate spacer **GS1** or the gate spacer **GS3**. In some embodiments, the gate spacer **GS1**, the gate spacer **GS2**, and the gate spacer **GS3** respectively have a thickness of from about 5 Å to about 1000 Å.

(23) In the operation **107** of FIG. **1**, a source and a drain is formed adjacent to the gate spacers. Referring to FIG. **12**, a source **S** and a drain **D** are formed adjacent to the gate spacers **GS1**, **GS2**, **GS3**. In some embodiments, the source **S** and the drain **D** are formed by the steps described below. The exposed portions of the fin **250** are partially removed (or partially recessed) to form recesses. Subsequently, one or more epitaxy or epitaxial (epi) processes are performed to form the source **S** and the drain **D**. In examples, the source **S** and the drain **D** may include Si features, SiGe features, silicon phosphate (SiP) features, silicon carbide (SiC) features and/or other suitable features with a crystalline. For example, the source **S** and the drain **D** may include GaAs, GaP, GaN, InP, InAs, InSb, GaAsP, AlGaAs, AlInAs, AlGaAs, GaInAs, GaInP, GaInAsP, or a combination thereof.

(24) In the operation **109** of FIG. **1**, a contact etch stop layer and an interlayer dielectric layer are formed adjacent to the gate spacers. Turning now to FIG. **13**, in some embodiments, a contact etch stop layer (CESL) **310** is formed adjacent to the gate spacer **GS3**. The interlayer dielectric layer (ILD) **320** is formed adjacent to the CESL **310**. More specifically, the CESL **310** is interposed between the gate spacer **GS3** and the interlayer dielectric layer (ILD) **320**. In some embodiments, the CESL **310** and the ILD layer **320** are formed by the steps described below. A CESL material layer is blanket deposited covering the structure shown in FIG. **12**, and then an ILD material layer is blanket deposited covering the CESL. Afterwards, a CMP process may be performed to remove excessive portions of the ILD material layer and the CESL material layer to expose the dummy gate structure **DG**. In some embodiments, the CMP process may planarize the top surface of the

ILD layer **320** with the top surfaces of the dummy gate structure DG. In example, the CESL **310** may include a dielectric material, such as SiN, SiON, SiCON, SiC, SiOC, SiC_xN_y, SiO_x, or a combination thereof.

(25) In some embodiments, the CESL material layer may be deposited using plasma enhanced chemical vapor deposition (PECVD), sub atmospheric chemical vapor deposition (SACVD), low pressure chemical vapor deposition (LPCVD), atomic layer deposition (ALD), high-density plasma (HDP), plasma enhanced atomic layer deposition (PEALD), molecular layer deposition (MLD), or plasma impulse chemical vapor deposition (PICVD). In some embodiments, the CESL **310** has a thickness of from about 5 Å to about 500 Å.

(26) In some embodiments, the ILD material layer may be formed by CVD, ALD, spin-on-glass (SOG) or other suitable techniques. In some embodiments, the ILD layer **320** includes a material different from that of the CESL **310**. For example, the ILD layer **320** may include silicon oxide, silicon nitride, silicon oxynitride, tetraethoxysilane (TEOS), phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), low-k dielectric material, and/or other suitable dielectric materials. Examples of low-k dielectric materials include, but are not limited to, fluorinated silica glass (FSG), carbon doped silicon oxide, amorphous fluorinated carbon, parylene, bis-benzocyclobutenes (BCB), or polyimide.

(27) In the operation **111** of FIG. **1**, the dummy gate structure is replaced with a gate structure. Attention is now invited to FIG. **14**. The dummy gate structure DG is replaced with a gate structure G crossing over the fin **250** of the semiconductor substrate **210**. The gate structure G includes a gate dielectric layer **362**, a barrier layer **364**, and a work function conductor layer **366**. In some embodiments, the gate structure G is a high-k metal gate (HKMG), in which the gate dielectric layer **362** is made of a high-k dielectric material.

(28) In some embodiments, the gate structure G may be formed by the steps described below. The dummy gate structure DG is removed to form a trench exposing the first gate spacer GS1. A layer of gate dielectric is conformally formed to cover the inner surface of the trench. In some embodiments, the layer of gate dielectric is formed using a deposition processes, such as CVD, PVD, ALD, plasma enhanced CVD (PECVD) or other suitable techniques. For example, the layer of gate dielectric is made of high-k dielectric materials, such as metal oxides, transition metal-oxides, or the like. Examples of the high-k dielectric material include, but are not limited to, hafnium oxide (HfO₂), hafnium silicon oxide (HfSiO), hafnium tantalum oxide (HfTaO), hafnium titanium oxide (HfTiO), hafnium zirconium oxide (HfZrO), zirconium oxide (ZrO₂), zirconium silicon oxide (ZrSiO₄), lanthanum oxide (La₂O₃), titanium oxide (TiO₂), tantalum pentoxide (Ta₂O₅), strontium titanium oxide (SrTiO₃), aluminum oxide (Al₂O₃), hafnium dioxide-alumina (HfO₂—Al₂O₃) alloy, scandium oxide (Sc₂O₃), barium strontium titanium oxide (Ba_{0.6}Sr_{0.4}TiO₃), yttrium oxide (Y₂O₃), a combination thereof, or other suitable dielectric materials. The layer of gate dielectric may include various dielectric materials, either individually or in combination.

(29) Thereafter, in some embodiments, a layer of barrier material is conformally formed on the layer of gate dielectric. In some embodiments, the layer of barrier material is formed using ALD, CVD, LPCVD or MLD. For example, the barrier material may include TiN, TaN, AlN, SiN, TiC, TaC, AlC, SiC or a combination thereof. In some embodiments, the layer of barrier material may be omitted.

(30) Thereafter, in some embodiments, a layer of work function conductor material is conformally formed on the barrier layer **364**. In some embodiments, the layer of work function conductor material includes W, Al, Cu, AlCu, Ti, TiAlN, TaC, TaCN, TaSiN, Mn, Zr, TiN, Ta, TaN, Co, Ni, or a combination thereof. In some embodiments, the layer of work function conductor material has a thickness of about 5 Å to about 500 Å. In some embodiments, the layer of work function conductor material includes a material different from that of the barrier layer **364**.

(31) Thereafter, a CMP is carried out to polish the deposited layers of work function conductor

material, barrier material, and the gate dielectric to obtain the structure depicted in FIG. 14.

(32) Attention is now invited to FIGS. 14, 15A, and 15B. FIG. 15A is a cross-sectional view along a line A-A' in FIG. 14. FIG. 15B is a cross-sectional view along a line B-B' in FIG. 14. The gate structure G is disposed on the semiconductor substrate 210. More specifically, the gate structure G has a part disposed on the fin 250 of the semiconductor substrate 210 as shown in FIG. 15A, and a part disposed on the isolation dielectric layer 260 as shown in FIG. 15B. The gate oxide layer 270 may be disposed between the gate structure G and the fin 250 as shown in FIG. 15A. Referring to FIGS. 15A and 15B simultaneously, the gate spacer GS1 abuts the sidewall GSW of the gate structure G. More specifically, the gate spacer GS1 has a vertical portion GS1V on the sidewall GSW of the gate structure G and a lateral portion GS1L that extends from a bottom of the vertical portion GS1V away from the gate structure G. In some embodiments, the gate spacer GS1 has an L-shape cross-section. The gate spacer GS2 is in contact with the gate spacer GS1, and has a vertical portion GS2V on a sidewall SW1 of the gate spacer GS1 and a lateral portion GS2L that extends from a bottom of the vertical portion GS2V away from the gate spacer GS1. In some embodiments, the gate spacer GS2 has an L-shape cross-section. The gate spacer GS3 is in contact with the gate spacer GS2. The gate spacer GS3 abuts a sidewall SW2 of the gate spacer GS2 and is disposed on the lateral portion GS2L of the gate spacer GS2. In some embodiments, the lateral portion GS1L of the gate spacer GS1 has an edge E aligned with the lateral portion GS2L of the gate spacer GS2 and the gate spacer GS3. The interlayer dielectric layer 320 is neighboring to the gate spacer GS1, the gate spacer GS2, and the gate spacer GS3. The CESL 310 is disposed between the interlayer dielectric layer 320 and the gate spacers GS1, GS2, GS3. More specifically, the CESL 310 is disposed between the interlayer dielectric layer 320 and each of the lateral portion GS1L of the gate spacer GS1, and the lateral portion GS2L of the gate spacer GS2.

(33) In operation 113 of FIG. 1, at least one portion of one of the gate spacers is removed.

Reference is made to FIGS. 16A-16B. A portion of the gate spacer GS2 is removed to form a gap G1 between the gate spacer GS1 and the gate spacer GS3. In other words, the gate spacer GS2 is partially removed to form the gap G1. The gate spacer GS2 may act as a sacrificial gate spacer. As shown in FIG. 16A, a portion of the lateral portion GS2L of the gate spacer GS2 remains under the gate spacer GS3 and over the lateral portion GS1L of the gate spacer GS1. The remained lateral portion GS2L is under the gate spacer GS3 and over the lateral portion GS1L of the gate spacer GS1. As shown in FIG. 16B, a portion of the vertical portion GS2V of the gate spacer GS2 over the isolation dielectric layer 260 remains between the gate spacer GS1 and the gate spacer GS3. In some embodiments, the gate spacer GS2 is removed by an etching process, such as dry etching. In the removing process, only the gate spacer GS2 is removed. In some embodiments, the gate spacer GS2 includes or consists of a material different from that of the gate spacer GS1 and/or the gate spacer GS3. In some embodiments, the gate spacer GS2 has an etch resistance different from that of the gate spacer GS1 and/or the gate spacer GS3. In some embodiments, the gate spacer GS2 has an etch resistance lower than that of the gate spacer GS1 and/or the gate spacer GS3. In some other embodiments, the gate spacer GS2 may be entirely removed (not shown in FIGS. 16A-16B).

(34) In the operation 115 of FIG. 1, a contact stop layer is formed over the gate structure, the gate spacers, and the interlayer dielectric layer to form an air gap. Attention is now invited to FIGS. 17A-17B. A contact stop layer CS1 is formed over the gate structure G, the gate spacers GS1, GS2, GS3, the CESL 310, and the interlayer dielectric layer 320 to form an air gap AR1 present between the gate spacer GS1, the contact stop layer CS1, and the interlayer dielectric layer 320. As such, a semiconductor structure 1700 is formed. The gate spacer GS3 is disposed between the air gap AR1 and the interlayer dielectric layer 320. The CESL 310 is disposed between the gate spacer GS3 and the interlayer dielectric layer 320. The contact stop layer CS1 includes a capping portion CP1 that seals a top of the air gap AR1. The capping portion CP1 extends toward the top of the air gap AR1. The air gap AR1 contacts the gate spacers GS1, GS3 and the capping portion CP1 of the contact stop layer CS1, and optionally the remained gate spacer GS2. The gate spacers GS1, GS3 (and

optionally the remained gate spacer GS2) and the capping portion CP1 of the contact stop layer CS1 respectively constitute a portion of a boundary of the air gap AR1. In some embodiments, the contact stop layer CS1 further includes a horizontal layer HL1, and the capping portion CP1 extends downwards from the horizontal layer HL1 and to a position between the gate spacer GS1 and the gate spacer GS3 (or between the gate spacer GS1 and the interlayer dielectric layer 320). (35) Referring to FIGS. 17A-17B simultaneously, the air gap AR1 has a first air gap portion AR11 and a second air gap portion AR12 communicated with the first air gap portion AR11 over the gate oxide layer 270 as shown in FIG. 17A. The air gap AR1 further has a third air gap portion AR13 over the isolation dielectric layer 260, as shown in FIG. 17B, communicated with the first air gap portion AR11 and the second air gap portion AR12. The first air gap portion AR11 is present between the vertical portion GS1V of the gate spacer GS1 and the gate spacer GS3. The second air gap portion AR12 is sandwiched between the gate spacer GS3 and the lateral portion GS1L of the gate spacer GS1. In some embodiments, a height H1 between a lower surface LS of the capping portion CP1 and an upper surface US of the lateral portion GS1V of the gate spacer GS1 ranges from about 10 Å to about 1000 Å. In some embodiments, a distance D1 between the remained lateral portion GS2L and the sidewall SW1 of the gate spacer GS1 ranges from about 2 Å to about 200 Å. In some embodiments, the third air gap portion AR13 has a height H2 ranges from about 10 Å to about 800 Å. In some embodiments, the remained vertical portion GS2V of the gate spacer GS2 shown in FIG. 17B has a height H3 ranged from about 2 Å to about 200 Å.

(36) FIG. 18 is a cross-sectional view illustrating a semiconductor structure 1800, in accordance with yet some embodiments of the instant disclosure. The difference between the semiconductor structure 1800 shown in FIG. 18 and the semiconductor structure 1700 shown in FIG. 17A is that, after the operation 113, a portion of the vertical portion GS2V of the gate spacer GS2 of FIG. 18 remains between the gate spacer GS1 and the gate spacer GS3. Further, a portion of the lateral portion GS2L remains under the gate spacer GS3 and over the lateral portion GS1L of the gate spacer GS1. However, the gate spacer GS2 of FIG. 17A does not have the remained vertical portion GS2V between the gate spacer GS1 and the gate spacer GS3. As shown in FIG. 18, an air gap AR1' is disposed between the gate spacers GS1 and GS3 and between the gate spacer GS2 and the capping portion CP1 of the contact stop layer CS1, and contacts the gate spacers GS1, GS2, GS3 and the capping portion CP1 of the contact stop layer CS1.

(37) Reference is made to FIGS. 19A-19B, which are cross-sectional views illustrating the structure after the operation 113 of FIG. 1 is performed, according to yet some embodiments. A portion of the gate spacer GS3 is removed to form a gap G2 between the gate spacer GS2 and the CESL 310. The gate spacer GS3 may act as a sacrificial gate spacer. As shown in FIG. 19A, the portion of the gate spacer GS3 over the gate oxide layer 270 is entirely removed. As shown in FIG. 19B, however, the portion of the gate spacer GS3 over the isolation dielectric layer 260 remains between the gate spacer GS2 and the CESL 310. In yet some other embodiments, the portion of the gate spacer GS3 over the isolation dielectric layer 260 may be entirely removed (not shown in FIG. 19B). In some embodiments, the gate spacer GS3 is removed by an etching process, such as dry etching. In the removing operation shown in FIGS. 19A and 19B, only the gate spacer GS3 is removed. In some embodiments, the gate spacer GS3 includes or consists of a material different from that of the gate spacer GS2 and/or the CESL 310. In some embodiments, the gate spacer GS3 has an etch resistance different from that of the gate spacer GS2 and/or the CESL 310. In some embodiments, the gate spacer GS3 has an etch resistance lower than that of the gate spacer GS2 and/or the CESL 310.

(38) Attention is now invited to FIGS. 20A-20B, that are cross-sectional views illustrating the structure after the operation 115 of FIG. 1 is performed, according to yet some embodiments. A contact stop layer CS2 is formed over the gate structure G, the gate spacers GS1, GS2, the remained gate spacer GS3, the CESL 310, and the interlayer dielectric layer 320 to form an air gap AR2 present between the gate spacer GS1, the contact stop layer CS2, and the interlayer dielectric

layer **320**. As such, a semiconductor structure **2000** is formed. The gate spacer **GS2** is disposed between the gate spacer **GS1** and the air gap **AR2**. The CESL **310** is disposed between the gate spacers **GS1**, **GS2** and the interlayer dielectric layer **320**. The contact stop layer **CS2** includes a capping portion **CP2** that seals a top of the air gap **AR2**. The capping portion **CP2** extends toward the top of the air gap **AR2**. The air gap **AR2** contacts the gate spacer **GS2**, the remained spacer **GS3**, the capping portion **CP2** of the contact stop layer **CS2**, and the CESL **310**. The gate spacer **GS2**, the remained spacer **GS3**, the capping portion **CP2** of the contact stop layer **CS2**, and the CESL **310** respectively constitute a portion of a boundary of the air gap **AR2**. In some embodiments, the contact stop layer **CS2** further includes a horizontal layer **HL2**, and the capping portion **CP2** extends downwards from the horizontal layer **HL2** and to a position between the gate spacer **GS2** and the CESL **310** (or between the gate spacer **GS1** and the interlayer dielectric layer **320**).

(39) Referring to FIGS. **20A-20B** simultaneously, the air gap **AR2** has a first air gap portion **AR21** on the gate oxide layer **270** as shown in FIG. **20A** and a second air gap portion **AR22** communicated with the first air gap portion **AR21** on the isolation dielectric layer **260** as shown in FIG. **20B**. The first air gap portion **AR21** is present between the vertical portion **GS2V** of the gate spacer **GS2** and the CESL **310**, and directly over the lateral portion **GS2L** of the gate spacer **GS2**. The lateral portion **GS2L** of the gate spacer **GS2** is under the first air gap portion **AR21** of the air gap **AR2**. In some embodiments, the first air gap portion **AR21** has a height **H4** ranges from about 10 Å to about 800 Å. As shown in FIG. **20B**, remained portion of the gate spacer **GS3** is under the second air gap portion **AR22**. In some embodiments, the second air gap portion **AR22** has a height **H5** ranged from about 10 Å to about 600 Å. In some embodiments, the remained gate spacer **GS3** shown in FIG. **20B** has a height **H6** of from about 2 Å to about 150 Å.

(40) FIG. **21** is a cross-sectional view illustrating a semiconductor structure **2100**, in accordance with yet some embodiments of the instant disclosure. The difference between the semiconductor structure **2100** shown in FIG. **21** and the semiconductor structure **2000** shown in FIG. **20A** is that, after the operation **113**, a portion of the gate spacer **GS3** of FIG. **21** over the gate oxide layer **270** remains between the gate spacer **GS2** and the CESL **310** and over the lateral portion **GS2L** of the gate spacer **GS2**. However, FIG. **20A** shows the portion of gate spacer **GS3** over the gate oxide layer **270** is entirely removed. As shown in FIG. **21**, the remained portion of the gate spacer **GS3** is under an air gap **AR2'**. The air gap **AR2'** is located between the gate spacer **GS2** and the CESL **310**, and between the remained gate spacer **GS3** and the capping portion **CP1** of the contact stop layer **CS1**. The air gap **AR2'** contacts the gate spacer **GS2**, the remained gate spacer **GS3**, the CESL **310** and the capping portion **CP2** of the contact stop layer **CS2**.

(41) In yet some other embodiments, the method **100** may further include removing at least one portion of the contact etch stop layer. For example, the removal of the at least one portion of the contact etch stop layer may be performed before or after the operation **113**. Alternatively, the removing the at least one portion of the contact etch stop layer and the operation **113** are performed concurrently. Reference is made to FIGS. **22A-22B**. A portion of the gate spacer **GS3** and a portion of the CESL **310** are removed to form a gap **G3** between the gate spacer **GS2** and the ILD layer **320**. As shown in FIG. **22A**, the gate spacer **GS3** over the gate oxide layer **270** is entirely removed, and the CESL **310** is partially removed. As shown in FIG. **22B**, the gate spacer **GS3** and the CESL **310** on the isolation dielectric layer **260** are partially removed. A portion of the gate spacer **GS3** remains over the lateral portion **GS2L** of the gate spacer **GS2** and a portion of the CESL **310** remains next to the remained portion of the gate spacer **GS3**. In some embodiments, the gate spacer **GS3** and the CESL **310** are removed by etching processes, such as dry etching processes. In some embodiments, the gate spacer **GS3** includes or consists of a material different from that of the gate spacer **GS2** and/or the CESL **310**. In some embodiments, the gate spacer **GS3** has an etch resistance different from that of the gate spacer **GS2** and/or the CESL **310**. In some embodiments, the gate spacer **GS3** has an etch resistance lower than that of the gate spacer **GS2** and/or the CESL

310. In some embodiments, the CESL **310** includes or consists of a material different from that of the interlayer dielectric layer **320**. In some embodiments, the CESL **310** has an etch resistance different from that of the interlayer dielectric layer **320**. In some embodiments, the CESL **310** has an etch resistance lower than that of the interlayer dielectric layer **320**.

(42) FIGS. **23A-23B** are cross-sectional views illustrating the structure after the operation **115** of FIG. **1** is performed, according to yet some embodiments. A contact stop layer **CS3** is formed over the gate structure **G**, the gate spacers **GS1**, **GS2**, the remained gate spacer **GS3**, the remained CESL **310**, and the interlayer dielectric layer **320** to form an air gap **AR3** present between the gate spacer **GS1**, the contact stop layer **CS3**, and the interlayer dielectric layer **320**. As such, a semiconductor structure **2300** is formed. The gate spacer **GS2** is disposed between the gate spacer **GS1** and the air gap **AR3**. The remained CESL **310** is disposed between the gate spacers **GS1**, **GS2** and the interlayer dielectric layer **320**. The contact stop layer **CS3** includes a capping portion **CP3** that seals a top of the air gap **AR3**. The capping portion **CP3** extends toward the top of the air gap **AR3**. The air gap **AR3** contacts the gate spacer **GS2**, the remained gate spacer **GS3**, the capping portion **CP3** of the contact stop layer **CS3**, the remained CESL **310**, and the interlayer dielectric layer **320**. The gate spacer **GS2** and the remained gate spacer **GS3**, the capping portion **CP3** of the contact stop layer **CS3**, the remained CESL **310**, and the interlayer dielectric layer **320** respectively constitute a portion of a boundary of the air gap **AR3**. In some embodiments, the contact stop layer **CS3** further includes a horizontal layer **HL3**, and the capping portion **CP3** extends downwards from the horizontal layer **HL3** and to a position between the gate spacer **GS2** and the interlayer dielectric layer **320** (or between the gate spacer **GS1** and the interlayer dielectric layer **320**).

(43) Please referring to FIGS. **23A-23B** simultaneously, the air gap **AR3** has a first air gap portion **AR31** over the gate oxide layer **270** as shown in FIG. **23A** and a second air gap portion **AR32** over the isolation dielectric layer **260** that communicates with the first air gap portion **AR31**, as shown in FIG. **23B**. The first air gap portion **AR31** is present between the vertical portion **GS2V** of the gate spacer **GS2** and the interlayer dielectric layer **320**, and over the lateral portion **GS2L** of the gate spacer **GS2** and the CESL **310**. The lateral portion **GS2L** is under the air gap **AR3**. In some embodiments, the first air gap portion **AR31** has a height **H7** is of from about 10 Å to about 600 Å. As shown in FIG. **23B**, portions of the gate spacer **GS3** and the CESL **310** are disposed under the second air gap portion **AR32**. In some embodiments, the second air gap portion **AR32** has a height **H8** of from about 10 Å to about 600 Å. In some embodiments, the remaining gate spacer **GS3** shown in FIG. **23B** has a height **H9** of from about 2 Å to about 150 Å.

(44) FIG. **24** is a cross-sectional view illustrating a semiconductor structure **2400**, in accordance with some embodiments of the instant disclosure. The difference between the semiconductor structure **2400** shown in FIG. **24** and the semiconductor structure **2300** shown in FIG. **23A** is that, after removing a portion of the gate spacer **GS3** and a portion of the CESL **310**, a portion of the gate spacer **GS3** in FIG. **24** over the gate oxide layer **270** remains between the gate spacer **GS2** and the CESL **310**, and a portion of the CESL **310** in FIG. **24** over the gate oxide layer **270** remains between the gate spacer **GS3** and the ILD layer **320**. As shown in FIG. **24**, the remained portion of the gate spacer **GS3** and the remained portion of the CESL **310** are under an air gap **AR3'**. The air gap **AR3'** is disposed between the gate spacer **GS2** and the ILD layer **320**, and between the gate spacer **GS3**, the CESL **310**, and the capping portion **CP1** of the contact stop layer **CS1**. The air gap **AR3'** contacts the gate spacer **GS2**, remained gate spacer **GS3**, the remained CESL **310**, the capping portion **CP2** of the contact stop layer **CS2**, and the ILD layer **320**.

(45) For a semiconductor structure, such as fin effect transistor (FinFET), lowering the parasitic capacitance between the gates and the drains or between the gates and the sources is good for reducing the gate pitches. The instant disclosure provides the air gaps adjacent to the gate structures in various embodiments to lower the parasitic capacitance and to mitigate current leakage.

Accordingly, the performance of the semiconductor structure is promoted.

(46) According to some embodiments, a device includes a semiconductive fin having source and

drain regions and a channel region between the source and drain regions, a gate feature over the channel region of the semiconductive fin, a first spacer around the gate feature, source and drain features respectively in the source and drain regions of the semiconductive fin, an interlayer dielectric layer around the first spacer, and a void between the first spacer and the interlayer dielectric layer and spaced apart from the gate feature and the source and drain features.

(47) According to some embodiments, a device includes a semiconductive fin having source and drain regions and a channel region between the source and drain regions, a gate feature over the channel region of the semiconductive fin, a first spacer around the gate feature, a second spacer around the first spacer, a void between the first spacer and the second spacer, and source and drain features respectively in the source and drain regions of the semiconductive fin.

(48) According to some embodiments, a device includes a semiconductive fin having source and drain regions and a channel region between the source and drain regions, a gate feature over the channel region of the semiconductive fin, a first spacer around the gate feature, source and drain features respectively in the source and drain regions of the semiconductive fin, an interlayer dielectric layer around the first spacer, a void between the first spacer and the interlayer dielectric layer, and a contact stop layer. The contact stop layer has a first portion directly over the gate feature, a second portion directly over the first spacer, and a third portion directly over the void, in which a bottom surface of the third portion of the contact stop layer is lower than a top surface of the first spacer.

(49) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A device, comprising: a transistor channel region; a transistor gate over the transistor channel region; and a multilayer gate spacer covering a sidewall of the transistor gate, the multilayer gate spacer comprising: a first spacer layer on the sidewall of the transistor gate; a second spacer layer on an outermost sidewall of the first spacer layer; a third spacer layer on an outermost sidewall of the second spacer layer, wherein when viewed in a cross-sectional view taken along a direction perpendicular to a longitudinal axis of the transistor gate, the third spacer layer has a cross-sectional pattern having a different shape than cross-sectional patterns of the first and second spacer layers, and the cross-sectional pattern of the third spacer layer has a top lower than tops of the cross-sectional patterns of the first and second spacer layers, wherein a bottom of the cross-sectional pattern of the third spacer layer has a horizontal dimension less than a vertical dimension of an outermost side of the cross-sectional pattern of the third spacer layer; an air-filled region atop the third spacer layer; a source/drain region adjacent to the transistor channel region; and a contact etch stop layer over the source/drain region, wherein the contact etch stop layer defines a vertical boundary of the air-filled region, and the contact etch stop layer has a top end level with or higher than a top end of the first spacer layer.
2. The device of claim 1, wherein the cross-sectional pattern of the third spacer layer is linear-shaped.
3. The device of claim 1, wherein one or more of the cross-sectional patterns of the first and second spacer layers are L-shaped.
4. The device of claim 1, wherein the top of the cross-sectional pattern of the third spacer layer is a

bottom boundary of an air gap.

5. A device, comprising: a semiconductor pedestal over a substrate; a gate structure across the semiconductor pedestal; a multilayer spacer alongside the gate structure, the multilayer spacer comprising: a first spacer layer; a second spacer layer on an outermost sidewall of the second spacer layer; and a third spacer layer on an outermost sidewall of the second spacer layer; a capping layer over the multilayer spacer, the capping layer and the third spacer layer defining an air gap between a top of the third spacer layer and a bottom of the capping layer, wherein the bottom of the capping layer in the air gap vertically overlaps with the third spacer layer; a source/drain region over the semiconductor pedestal; and a contact etch stop layer over the source/drain region, wherein the contact etch stop layer defines a vertical boundary of the air gap, and the contact etch stop layer has a top end level with or higher than a top end of the first spacer layer.

6. The device of claim 5, wherein the air gap has a top boundary lower than a top end of first spacer layer.

7. The device of claim 5, wherein the air gap has a top boundary lower than a top end of the second spacer layer.

8. The device of claim 5, wherein the air gap has a height greater than a height of the third spacer layer.

9. The device of claim 5, wherein the top of the third spacer layer is lower than a top of the second spacer layer.

10. The device of claim 5, wherein the top of the third spacer layer is lower than a top of the first spacer layer.

11. A device, comprising: a semiconductor fin extending from a substrate; a gate structure extending across the semiconductor fin; and a multilayer gate spacer abutting a sidewall of the gate structure, the multilayer gate spacer comprising: a first spacer layer in contact with the sidewall of the gate structure; a second spacer layer in contact with a sidewall of the first spacer layer oriented away from the gate structure; a third spacer layer in contact with a sidewall of the second spacer layer oriented away from the second spacer layer; and an air-filled region atop the third spacer layer; a source/drain region over the semiconductor fin; and a contact etch stop layer over the source/drain region, wherein the contact etch stop layer defines a vertical boundary of the air-filled region, and the contact etch stop layer has a top end not lower than a top end of the first spacer layer.

12. The device of claim 11, further comprising: a dielectric layer defining a top boundary of the air-filled region.

13. The device of claim 11, wherein the air-filled region has a top boundary lower than a top of the second spacer layer.

14. The device of claim 11, wherein the air-filled region has a top boundary lower than a top of the first spacer layer.

15. The device of claim 11, wherein the second spacer layer defines another vertical boundary of the air-filled region.

16. The device of claim 11, wherein the third spacer layer has a linear pattern in a cross-sectional view.

17. The device of claim 11, wherein the first spacer layer has an L-shaped pattern in a cross-sectional view.

18. The device of claim 11, wherein the second spacer layer has an L-shaped region in a cross-sectional view.

19. The device of claim 11, wherein the third spacer layer includes a material different from a material of the second spacer layer and a material of the first spacer layer.

20. The device of claim 1, wherein the horizontal dimension of the bottom of the cross-sectional pattern of the third spacer layer is the same as a horizontal dimension of the top of the cross-sectional pattern of the third spacer layer.

